

Compact magnetometer project update

- The lock-in workflow is still the same:
ODMR sweep $\rightarrow$ 16 parked frequencies
 $\rightarrow \Delta f(t) \rightarrow \Delta B(t)$.
- May 14 update: spike rejection is now active in live lock-in and live B-field acquisition.
- Latest parking run: 8 kept transitions, 16 parked frequencies, no rejected fitted centers.
- Live lock-in test: 266 batches in 21.3 s, about 12.5 Hz update rate.
- Offline B reconstruction: 266/266 samples rank-ok, condition number 9.74.
- Live B-field run: 387 batches in 60.0 s, about 6.4 Hz update rate.

Item	Current result
Transitions	8
Parked points	16
Single batch	113 ms
Live lock-in	12.5 Hz
Live B-field	6.4 Hz
B matrix cond.	9.74

Latest live lock-in spike replacements:

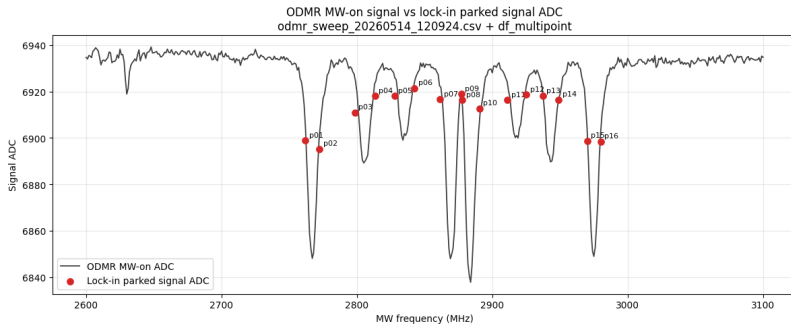
Signal cells	48
Reference cells	24

Workflow 1: choose 16 parked frequencies



- Latest ODMR sweep: May 14 12:09 run.
- The new parking algorithm kept all 8 detected transitions.
- Two parked frequencies are placed on each transition.
- The estimated bias field from the ODMR fit is approximately $(2.12, -0.59, -3.42)$ mT.

Workflow 2: acquire all parked points



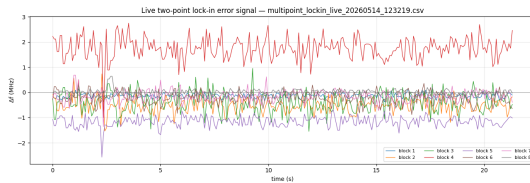
- One FPGA program measures all 16 parked frequencies.
- The one-batch reference is used as the baseline for live despiking.
- Latest one-batch acquisition took 113 ms.
- The predicted FPGA work per live batch is about 69 ms for 10 reps.

Workflow 3: convert lock-in signal to frequency shift

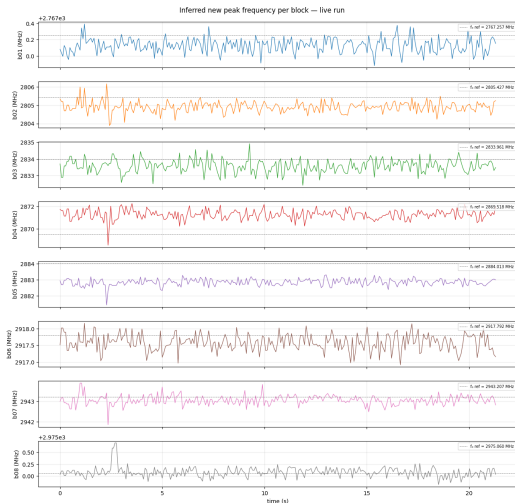
For each transition, use the two flanks:

$$\Delta f_k(t) = \frac{[S_+(t) - S_-(t)] - [S_{+,0} - S_{-,0}]}{m_- - m_+}.$$

- S_- and S_+ are the live signals at the lower and upper parked points.
- $S_{-,0}$ and $S_{+,0}$ come from the reference ODMR calibration.
- Latest live run: 266 batches over 21.3 s.
- Per-block frequency noise: 0.09 to 0.45 MHz standard deviation.
- Projected-field noise: 3.2 to 16.6 μT standard deviation.

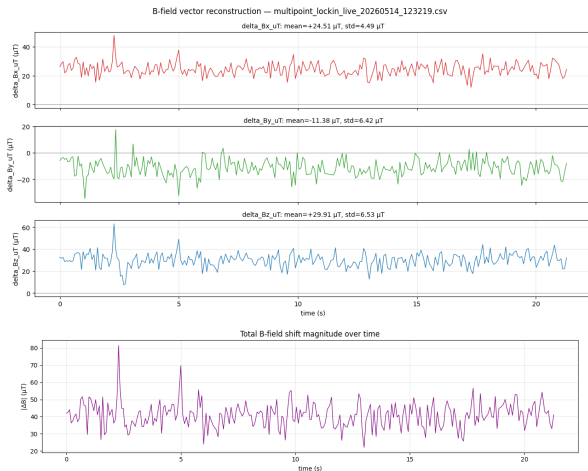


Workflow 4: inferred new peak frequencies



- This is the inferred new peak frequency for each of the 8 transitions.
- Solid traces show $f_{\text{new}}(t) = f_0 + \Delta f(t)$.
- The May 14 live run uses the causal Hampel despiker before the frequency conversion.
- This view is still useful for spotting remaining jumps before B reconstruction.

Workflow 5: reconstruct magnetic field



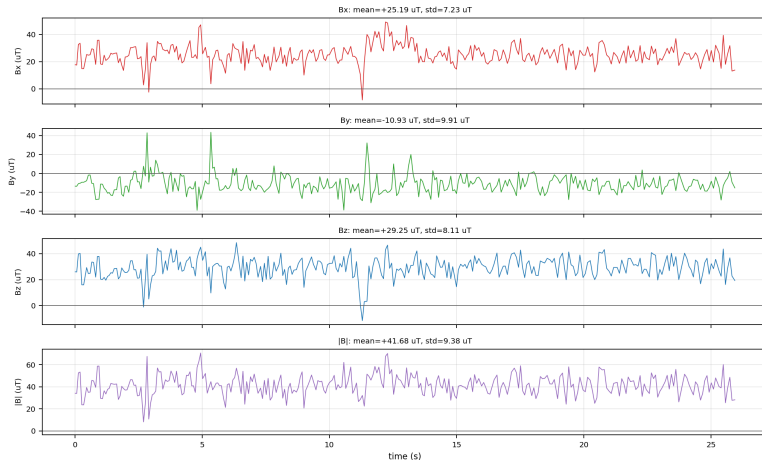
- Offline reconstruction of the May 14 no-magnet run.
- 266 samples, 266 rank-ok.
- B matrix condition number: 9.74.
- Reconstructed B statistics:

Comp.	Mean	Std.
ΔB_x	24.51	4.49
ΔB_y	-11.38	6.42
ΔB_z	29.91	6.53
$ \Delta B $	40.91	7.41

Units: μ T

New data: small magnet

Small magnet: reconstructed magnetic field - multipoint_lockin_live_20260514_121325_vector_rows.csv



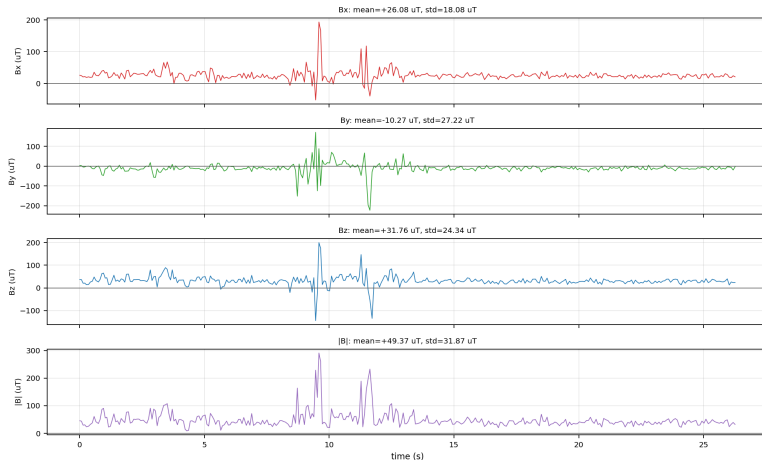
- Source: May 14 small-magnet run.
- 318 samples over 25.9 s.
- Live lock-in rate: 12.3 Hz.
- B matrix condition number: 9.74.

Comp.	Mean	Std.
ΔB_x	25.19	7.23
ΔB_y	-10.93	9.91
ΔB_z	29.25	8.11
$ \Delta B $	41.68	9.38

Units: μT

New data: neodymium big magnet

Neodymium big magnet: reconstructed magnetic field - multipoint_lockin_live_20260514_121149_vector_rows.csv

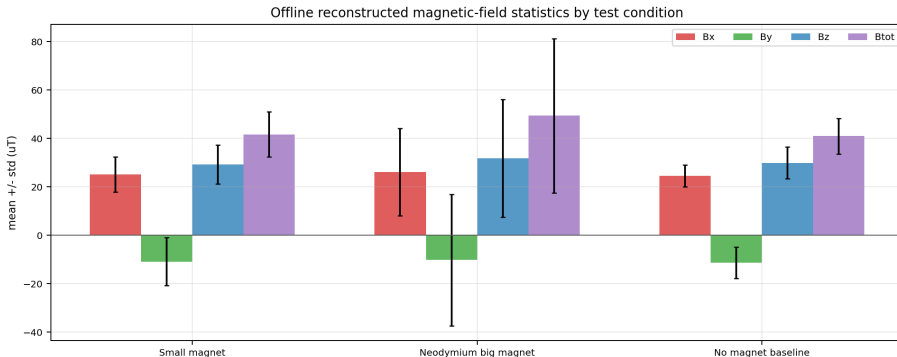


- Source: May 14 neodymium run.
- 344 samples over 26.3 s.
- Live lock-in rate: 13.1 Hz.
- This run shows larger transient spikes than the small-magnet run.

Comp.	Mean	Std.
ΔB_x	26.08	18.08
ΔB_y	-10.27	27.22
ΔB_z	31.76	24.34
$ \Delta B $	49.37	31.87

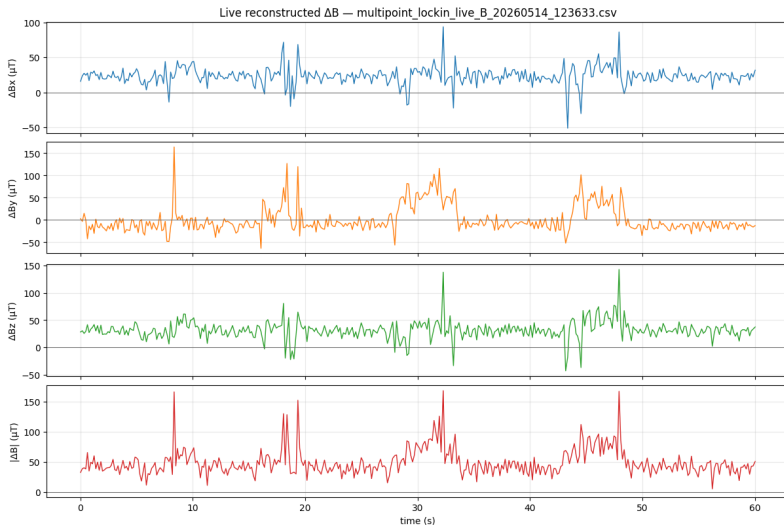
Units: μT

New data: offline comparison



- The comparison includes small magnet, neodymium big magnet, and no-magnet baseline.
- The neodymium run has the largest standard deviation because of transient jumps.
- Reporting both mean and standard deviation helps separate field offset from stability.

Workflow 6: live B-field measurement

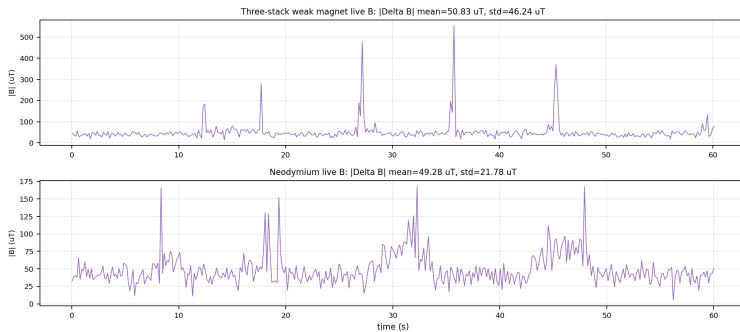


- Source: May 14 live-B run.
- 387 batches in 60.0 s.
- Update rate: about 6.4 Hz.
- 387/387 B reconstructions rank-ok.
- Spike replacements: 415 signal, 404 reference.

Comp.	Mean	Std.
ΔB_x	23.72	12.76
ΔB_y	0.77	29.79
ΔB_z	31.60	17.14
$ \Delta B $	49.28	21.78

Units: μT

Live B magnet comparison



Run	Mean	Std.
3-stack	50.83	46.24
Neodymium	49.28	21.78

Units: μT for $|\Delta B|$

- Both live-B test runs operated at about 6.5 Hz.
- The 3-stack run has the larger variation.
- Next step is to compare these live-B outputs against a known field geometry.

Next steps

Validate B reconstruction

- Apply a known test field and compare recovered direction and magnitude.
- Repeat no-magnet, small-magnet, and neodymium measurements with fixed geometry.
- Track residual norm, matrix rank, and condition number for every batch.
- Check whether the spike rejection removes artifacts without hiding real field changes.

Improve live operation

- Move live B closer to the 12.5 Hz raw live lock-in rate.
- Decouple acquisition, reconstruction, and plotting.
- Continue dual-channel measurement integration.
- Continue hardware optimization and patent manuscript documentation.